



Cerebrospinal fluid poly-ADP-ribose assay

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Summary (or Abstract)

We previously demonstrated using two discovery cohorts that cerebrospinal fluid (CSF) poly-ADP-ribose (PAR) levels were different amongst individuals with Parkinson's disease (PD) compared to controls (1). This investigation then sought to validate PAR in the CSF as a possible biomarker for PD diagnosis and progression using the PPMI cohort by assessing CSF PAR levels in both individuals with PD and controls at baseline and followed longitudinally. We obtained baseline and follow up CSF from PPMI participants and performed our sandwich ELISA assay to determine CSF PAR levels.

Method

Our CSF PAR assay is a sandwich ELISA that has been rigorously tested and optimized. Two different clones of monoclonal anti-PAR antibody were used, with the first antibody coated on the 96-well microtiter plate and the biotinylated anti-PAR antibody added after the PD patient or control patient CSF were added. The CSF was not diluted. We generated these human PAR antibodies that are specific to PAR, but not to other negative charged molecules including adenosin 5'-disphosphoribose (ADP), genomic DNA (gDNA), or RNA. Following incubation, the color change was detected with the HRP-conjugated streptavidin antibody.

References

1. Kam, T.I., et al., *Poly(ADP-ribose) drives pathologic alpha-synuclein neurodegeneration in Parkinson's disease*. Science, 2018. **362**(6414).

About the Authors

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